



Technical Service Bulletin

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X15 CM2350 X114B - Efficiency Series and X15 CM2350 X116B - Performance Series Engine Introduction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This engine introduction announces the X15 CM2350 X114B - Efficiency Series and X15 CM2350 X116B - Performance Series engines. These engines were designed for use in on-highway applications and complies with:

- Environmental Protection Agency (EPA)
 - 2013 Emission Levels
 - 2017 Greenhouse Gas Requirements
- Air Resources Board (ARB)
 - 2013 Emission Levels
- Diagnostic Requirements
 - On Board Diagnostics (OBD)

Engine Features

These engines replace the ISX15 CM2350 X101 engine.

See the following for a complete list of engine features in the Owners Manual, Bulletin 5411183. Refer to Procedure 100-006 in Section E (</qs3/pubsys2/xml/en/procedures/510/510-100-006.html>).

- Engine family names have been updated:
 - X1 – Efficiency Series
 - X3 – Performance Series
 - X2 – Family removed, the Efficiency and Performance horsepower (hp) ratings cover the previous X2 hp ratings.
- Compatible with Cummins® Connected Calibrations. Capable of over-the-air engine control module (ECM) calibration download and installation
- Exhaust System:

- Single Module™ aftertreatment; both the aftertreatment diesel particulate filter (DPF) and aftertreatment selective catalytic reduction (SCR) catalyst components are contained in one assembly
- UL2 diesel exhaust fluid (DEF) dosing system
- In-cylinder dosing and removal of the aftertreatment fuel injector
- Air Intake System:
 - Engine intake throttle actuator added to the Performance Series
 - Updated turbocharger – visibly smaller, simplified coolant plumbing
 - Fully integrated naturally aspirated air compressor
- Exhaust Gas Recirculation (EGR) System:
 - Improved EGR coolers
 - Different EGR cooler designs from the Performance to the Efficiency Series hardware
 - EGR venturi device replaced the previous orifice
- Fuel System:
 - Next Generation XPI fuel system introduced in the Performance Series
 - Updated – high-pressure fuel pump, fuel rail, and low-pressure fuel pump
 - Integrated thermal recirculating valve (TRV)
 - Using nylon low pressure fuel lines
- Lubricating Oil System:
 - Updated lubricating oil filter head and thermostat
 - Efficiency Series
 - Removed lubricating oil filter bypass
 - New lubricating oil pump
- Base Engine:
 - Diamond Like Coating (DLC) overhead pins
 - Updated cylinder head ports
 - Updated camshaft lobe profile
 - Increased compression ratio on the Efficiency Series (from 19:1 to 20:1)
 - New piston ring pack
 - Box structure piston
 - Oil pan design changes
 - Introducing a variable impactor style breather
 - Scissor gear removal
- Cooling System
 - Updated water pump and pulley

Recommendations and Specifications

Table 1, Recommendations and Specifications					
Manual Type	Bulletin	Procedure Title	Procedure (X15 CM2350 X114B)	Procedure (X15 CM2350 X116B)	Section
Operation and Maintenance Manuals	5411182 and 5411187	Lubricating Oil Recommendations and Specifications	Refer to Procedure 018-003 (/qs3/pubsyss2/xml/en/procedures/377/377-018-003.html)	Refer to Procedure 018-003 (/qs3/pubsyss2/xml/en/procedures/453/453-018-003.html)	V
		Fuel Recommendations and Specifications	Refer to Procedure 018-002 (/qs3/pubsyss2/xml/en/procedures/377/377-018-002.html)	Refer to Procedure 018-002 (/qs3/pubsyss2/xml/en/procedures/453/453-018-002.html)	
		Diesel Exhaust Fluid Recommendations and Specifications*	Refer to Procedure 018-026 (/qs3/pubsyss2/xml/en/procedures/99/99-018-026.html)	Refer to Procedure 018-026 (/qs3/pubsyss2/xml/en/procedures/99/99-018-026.html)	
		Coolant Recommendations and Specifications	Refer to Procedure 018-004 (/qs3/pubsyss2/xml/en/procedures/377/377-018-004.html)	Refer to Procedure 018-004 (/qs3/pubsyss2/xml/en/procedures/453/453-018-004.html)	
*For additional information on DEF. Reference Diesel Exhaust Fluid (DEF) Specifications for Cummins® Selective Catalytic Reduction (SCR) Systems, Bulletin 4021566.					
Maintenance Schedule					
Use the following procedures for maintenance intervals in the X15 CM2350 X114B and X15 CM2350 X116B Operation and Maintenance Manuals, Bulletins 5411182 and 5411187.					

- X15 CM2350 X114B: Refer to Procedure 359-001 in Section 3 (/qs3/pubsys2/xml/en/procedures/377/377-359-001.html).
- X15 CM2350 X116B: Refer to Procedure 359-001 in Section 3 (/qs3/pubsys2/xml/en/procedures/453/453-359-001.html).

Service Tools

New Cummins® service tools released to support the X15 CM2350 X114B and X15 CM2350 X116B engines are found in the Service Manuals, Bulletins 5411181 and 5411186.

For a complete list of Cummins® service tools needed to support this engine, consult a local Cummins® distributor.

INSITE™ Electronic Service Tool

INSITE™ electronic service tool version 8.0.2 or later supports full production engines. Version 8.0.2 or later includes all diagnostic tests and features necessary to fully support the X15 CM2350 X114B and X15 CM2350 X116B engines. INSITE™ electronic service tool updates are available through the Cummins® update manager. Reference the following website to access Cummins® update manager.

Note : <https://www.cummins.com/support/electronic-service-tools-support/insite-support> (<https://www.cummins.com/support/electronic-service-tools-support/insite-support>)

Expert Diagnostic System (EDS)

The EDS troubleshooting tool is available to fully support troubleshooting requirements for the X15 CM2350 X114B and X15 CM2350 X116B engines.

Engine Ratings

ST – SmartTorque2 Ratings

SA - SmartAdvantage™ Ratings

X15 CM2350 X114B - Efficiency Series – Line Haul/Regional Haul Truck

Model	Control Parts List (CPL)	Peak Torque [ft-lb at rpm]	Governed Speed [rpm]
X15 400	4342	1650 at 1000	1800
X15 400	4342	1750 at 1000	1800
X15 400	4342	1850 at 1000	1800
X15 400ST	4342	1450/1650 at 1000	1800
X15 400ST	4342	1550/1750 at 1000	1800
X15 400ST	4342	1650/1850 at 1000	1800
X15 400SA	4342	1450/1650 at 1000	1800
X15 400SA	4342	1550/1750 at 1000	1800
X15 400SA	4342	1550/1850 at 1000	1800
X15 450	4342	1650 at 1000	1800
X15 450	4342	1750 at 1000	1800

X15 CM2350 X114B - Efficiency Series – Line Haul/Regional Haul Truck			
Model	Control Parts List (CPL)	Peak Torque [ft-lb at rpm]	Governed Speed [rpm]
X15 450	4342	1850 at 1000	1800
X15 450ST	4342	1450/1650 at 1000	1800
X15 450ST	4342	1550/1750 at 1000	1800
X15 450ST	4342	1650/1850 at 1000	1800
X15 450SA	4342	1550/1750 at 1000	1800
X15 450SA	4342	1550/1850 at 1000	1800
X15 500	4342	1650 at 1000	1800
X15 500	4342	1850 at 1000	1800
X15 500ST	4342	1650/1850 at 1000	1800
X15 500SA	4342	1550/1850 at 1000	1800
X15 CM2350 X116B - Performance Series – Heavy Haul/Vocational Truck			
Model	Control Parts List (CPL)	Peak Torque [ft-lb at rpm]	Governed Speed [rpm]
X15 485	4343	1650 at 1150	2000
X15 485	4343	1850 at 1150	2000
X15 505	4343	1650 at 1150	2000
X15 505	4343	1850 at 1150	2000
X15 525	4343	1850 at 1150	2000
X15 565	4343	1850 at 1150	2000
X15 565	4343	2050 at 1150	2000
X15 605	4343	1850 at 1150	2000
X15 605	4343	2050 at 1150	2000
X15 CM2350 X116B - Performance Series – Emergency Vehicle (Must Comply to NFPA 1901/1906)			
Model	Control Parts List (CPL)	Peak Torque [ft-lb at rpm]	Governed Speed [rpm]
X15 505	4343	1850 at 1150	2100
X15 565	4343	1850 at 1150	2100
X15 600	4343	1850 at 1150	2100
X15 CM2350 X116B - Performance Series – Recreational Vehicle			
Model	Control Parts List (CPL)	Peak Torque [ft-lb at rpm]	Governed Speed [rpm]
X15 565	4343	1850 at 1150	2100
X15 605	4343	1950 at 1150	2100
Parts			

Reference the X15 CM2350 X114B and X15 CM2350 X116B Product Announcement, Bulletin AN1607, for parts information.

Additional Information

Table 3, Associated Procedures

Manual Type	Bulletin Number	Procedure Title	Procedure (X15 CM2350 X114B)	Procedure (X15 CM2350 X116B)	Section
Operation and Maintenance Manuals	5411182 and 5411187	Cummins® Product Technology	Refer to Procedure 100-006 (/qs3/pubsy s2/xml/en/pr ocedures/37 7/377-100-006.html)	Refer to Procedure 100-006 (/qs3/pubsy s2/xml/en/pr ocedures/45 3/453-100-006.html)	E
		Engine Identification	Refer to Procedure 100-001 (/qs3/pubsy s2/xml/en/pr ocedures/37 7/377-100-001.html)	Refer to Procedure 100-001 (/qs3/pubsy s2/xml/en/pr ocedures/45 3/453-100-001.html)	
		Engine Diagrams	Refer to Procedure 100-002 (/qs3/pubsy s2/xml/en/pr ocedures/37 7/377-100-002.html)	Refer to Procedure 100-002 (/qs3/pubsy s2/xml/en/pr ocedures/45 3/453-100-002.html)	
		General Engine	Refer to Procedure 018-015 (/qs3/pubsy s2/xml/en/pr ocedures/37 7/377-018-015.html)	Refer to Procedure 018-015 (/qs3/pubsy s2/xml/en/pr ocedures/45 3/453-018-015.html)	V

Table 3, Associated Procedures					
Manual Type	Bulletin Number	Procedure Title	Procedure (X15 CM2350 X114B)	Procedure (X15 CM2350 X116B)	Section
		Additional Service Literature	Refer to Procedure 205-001 (/qs3/pubsys2/xml/en/procedures/377/377-205-001.html)	Refer to Procedure 205-001 (/qs3/pubsys2/xml/en/procedures/453/453-205-001.html)	L

Document History

Date	Details
2016-11-3	Module Created
2018-10-4	Corrected URL.